Project Title: Campus Buddy: An Automated GUI Chatbot for College Campuses

MADE BY JAI RANE

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1. PROJECT SUBMISSION REPORT DETAILS

Student Credentials

1. **Project Title:** Campus Buddy: A Graphical UI Assistant for Student Onboarding
2. **Student Name:** Jai Rane
3. **Program**: Lenovo LEAP NextGen Scholar Program Internship
4. **Course / Batch:** Artificial Intelligence / June 2026
5. **Submission Date:** 13/05/2026

2. Project Objective & Description

- **Objective:** To streamline information access for newly admitted students and visiting parents during the hectic college admissions and orientation periods.

- **Description:** This project is a desktop-based, interactive graphical application designed to automate answers to high-frequency, repetitive queries. The chatbot provides immediate, automated assistance regarding admission criteria, fee timelines, library hours, hostel locations, and administrative helpline contacts. By using a clean button-driven dashboard, the application eliminates human typing errors and ensures smooth context navigation.

3. Technologies Used

- **Python 3.12:** The primary programming language utilized to write the backend state engine and conditional execution logic.
- **Tkinter Library:** Python's built-in interface framework used to design the layout, application windows, text boxes, and action buttons.
- **VS Code (Visual Studio Code):** The Integrated Development Environment (IDE) used to write, compile, and run the Python code.
- **Object-Oriented Event Handling:** Implemented via trigger buttons and lambda bindings to cleanly switch text outputs on user click actions.

4. Screenshots with Explanation

- 1. Code Screenshot 1

GUI based Chatbot.py

```
import tkinter as tk
from tkinter import scrolledtext

def process_choice(choice):
    # Clear the text box before inserting new information
    chat_display.config(state=tk.NORMAL)
    chat_display.delete('1.0', tk.END)

    if choice == 1:
        output_text = (
            "============================================\n"
            "📝 [ADMISSIONS INFO]:\n"
            "============================================\n"
            "• Entry Criteria: Minimum aggregate of 60% in high school.\n"
            "• Fee Deadline: The portal closes on the 15th of next month.\n"
            "• Status: Portals open for registration."
        )
    elif choice == 2:
        output_text = (
            "============================================\n"
            "🏢 [CAMPUS FACILITIES]:\n"
            "============================================\n"
            "• Central Library: Open daily from 8:00 AM to 9:00 PM.\n"
            "• Hostels: Student blocks are located on the North and South wings.\n"
            "• Wi-Fi: Available across campus for registered students."
        )
```

4. Screenshots with Explanation

- 1. Code Screenshot 2

```
        )
    elif choice == 3:
        output_text = (
            "==========================================\n"
            "📞 [EMERGENCY CONTACTS]:\n"
            "==========================================\n"
            "• Administrative Helpline: Call 022-245678.\n"
            "• Medical Emergency: Dial extension 104 on campus.\n"
            "• Security Desk: Available 24/7 at the front gates."
        )
    elif choice == 4:
        output_text = (
            "==========================================\n"
            "👋 CHATBOT TERMINATED\n"
            "==========================================\n"
            "Thank you for using Campus Buddy! Have a great day ahead! 🚀"
        )
        # Disable buttons upon exiting the chat app
        btn1.config(state=tk.DISABLED)
        btn2.config(state=tk.DISABLED)
        btn3.config(state=tk.DISABLED)
        btn4.config(state=tk.DISABLED)

    chat_display.insert(tk.END, output_text)
    chat_display.config(state=tk.DISABLED)

# Initialize Window Layout
window = tk.Tk()
window.title("Campus Buddy AI Chatbot")
window.geometry("500x450")
window.configure(bg="#1e1e2e")
```

4. Screenshots with Explanation

- 1. Code Screenshot 3

```python
# Title Header
title_label = tk.Label(window, text=" Campus Buddy Assistant", font=("Arial", 16, "bold"), bg="#1e1e2e", fg="#cdd6f4")
title_label.pack(pady=10)

# Welcome Sub-label
welcome_label = tk.Label(window, text="Hello! Click an option below to navigate your first days on campus.", font=("Arial", 10), bg="#1e1e2e", fg="#a6adc8")
welcome_label.pack(pady=5)

# Scrollable Output Display Panel
chat_display = scrolledtext.ScrolledText(window, width=55, height=12, font=("Courier New", 10), bg="#313244", fg="#a6e3a1", wrap=tk.WORD)
chat_display.insert(tk.END, "System Ready. Please select a panel below...")
chat_display.config(state=tk.DISABLED)
chat_display.pack(pady=10)

# Frame container for control options buttons
button_frame = tk.Frame(window, bg="#1e1e2e")
button_frame.pack(pady=10)

# Button Styling Array Configurations
btn_style = {"font": ("Arial", 10, "bold"), "bg": "#89b4fa", "fg": "#11111b", "width": 20, "pady": 5, "bd": 0, "activebackground": "#b4befe"}

btn1 = tk.Button(button_frame, text="1. Admissions Info", **btn_style, command=lambda: process_choice(1))
btn1.grid(row=0, column=0, padx=5, pady=5)

btn2 = tk.Button(button_frame, text="2. Campus Facilities", **btn_style, command=lambda: process_choice(2))
btn2.grid(row=0, column=1, padx=5, pady=5)

btn3 = tk.Button(button_frame, text="3. Emergency Contacts", **btn_style, command=lambda: process_choice(3))
btn3.grid(row=1, column=0, padx=5, pady=5)

btn4 = tk.Button(button_frame, text="4. Exit Assistant", font=("Arial", 10, "bold"), bg="#f38ba8", fg="#11111b", width=20, pady=5, bd=0, activebackground="#f5e0dc", command=lambda: process_choice(4))
```

4. Screenshots with Explanation

- 1. Code Screenshot 4

```
# Keep application window operational
window.mainloop()
import tkinter as tk
from tkinter import scrolledtext

def process_choice(choice):
    # Clear the text box before inserting new information
    chat_display.config(state=tk.NORMAL)
    chat_display.delete('1.0', tk.END)

    if choice == 1:
        output_text = (
            "==========================================\n"
            "📄 [ADMISSIONS INFO]:\n"
            "==========================================\n"
            "• Entry Criteria: Minimum aggregate of 60% in high school.\n"
            "• Fee Deadline: The portal closes on the 15th of next month.\n"
            "• Status: Portals open for registration."
        )
    elif choice == 2:
        output_text = (
            "==========================================\n"
            "🏢 [CAMPUS FACILITIES]:\n"
            "==========================================\n"
            "• Central Library: Open daily from 8:00 AM to 9:00 PM.\n"
            "• Hostels: Student blocks are located on the North and South wings.\n"
            "• Wi-Fi: Available across campus for registered students."
        )
    elif choice == 3:
        output_text = (
            "==========================================\n"
            "🚨 [EMERGENCY CONTACTS]:\n"
```

4. Screenshots with Explanation

- 1. Code Screenshot 5

```
    elif choice == 3:
        output_text = (
            "==========================================\n"
            "📞 [EMERGENCY CONTACTS]:\n"
            "==========================================\n"
            "• Administrative Helpline: Call 022-245678.\n"
            "• Medical Emergency: Dial extension 104 on campus.\n"
            "• Security Desk: Available 24/7 at the front gates."
        )
    elif choice == 4:
        output_text = (
            "==========================================\n"
            "👋 CHATBOT TERMINATED\n"
            "==========================================\n"
            "Thank you for using Campus Buddy! Have a great day ahead! 🚀"
        )
        # Disable buttons upon exiting the chat app
        btn1.config(state=tk.DISABLED)
        btn2.config(state=tk.DISABLED)
        btn3.config(state=tk.DISABLED)
        btn4.config(state=tk.DISABLED)

    chat_display.insert(tk.END, output_text)
    chat_display.config(state=tk.DISABLED)

# Initialize Window Layout
window = tk.Tk()
window.title("Campus Buddy AI Chatbot")
window.geometry("500x450")
window.configure(bg="#1e1e2e")

# Title Header
```

4. Screenshots with Explanation

- 1. Code Screenshot 6

```python
# Welcome Sub-label
welcome_label = tk.Label(window, text="Hello! Click an option below to navigate your first days on campus.", font=("Arial", 10), bg="#1e1e2e", fg="#a6adc8")
welcome_label.pack(pady=5)

# Scrollable Output Display Panel
chat_display = scrolledtext.ScrolledText(window, width=55, height=12, font=("Courier New", 10), bg="#313244", fg="#a6e3a1", wrap=tk.WORD)
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chat_display.config(state=tk.DISABLED)
chat_display.pack(pady=10)

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button_frame = tk.Frame(window, bg="#1e1e2e")
button_frame.pack(pady=10)

# Button Styling Array Configurations
btn_style = {"font": ("Arial", 10, "bold"), "bg": "#89b4fa", "fg": "#11111b", "width": 20, "pady": 5, "bd": 0, "activebackground": "#b4befe"}

btn1 = tk.Button(button_frame, text="1. Admissions Info", **btn_style, command=lambda: process_choice(1))
btn1.grid(row=0, column=0, padx=5, pady=5)

btn2 = tk.Button(button_frame, text="2. Campus Facilities", **btn_style, command=lambda: process_choice(2))
btn2.grid(row=0, column=1, padx=5, pady=5)

btn3 = tk.Button(button_frame, text="3. Emergency Contacts", **btn_style, command=lambda: process_choice(3))
btn3.grid(row=1, column=0, padx=5, pady=5)

btn4 = tk.Button(button_frame, text="4. Exit Assistant", font=("Arial", 10, "bold"), bg="#f38ba8", fg="#11111b", width=20, pady=5, bd=0, activebackground="#f5e0dc", command=lambda: process_choice(4))
btn4.grid(row=1, column=1, padx=5, pady=5)

# Keep application window operational
window.mainloop()
```
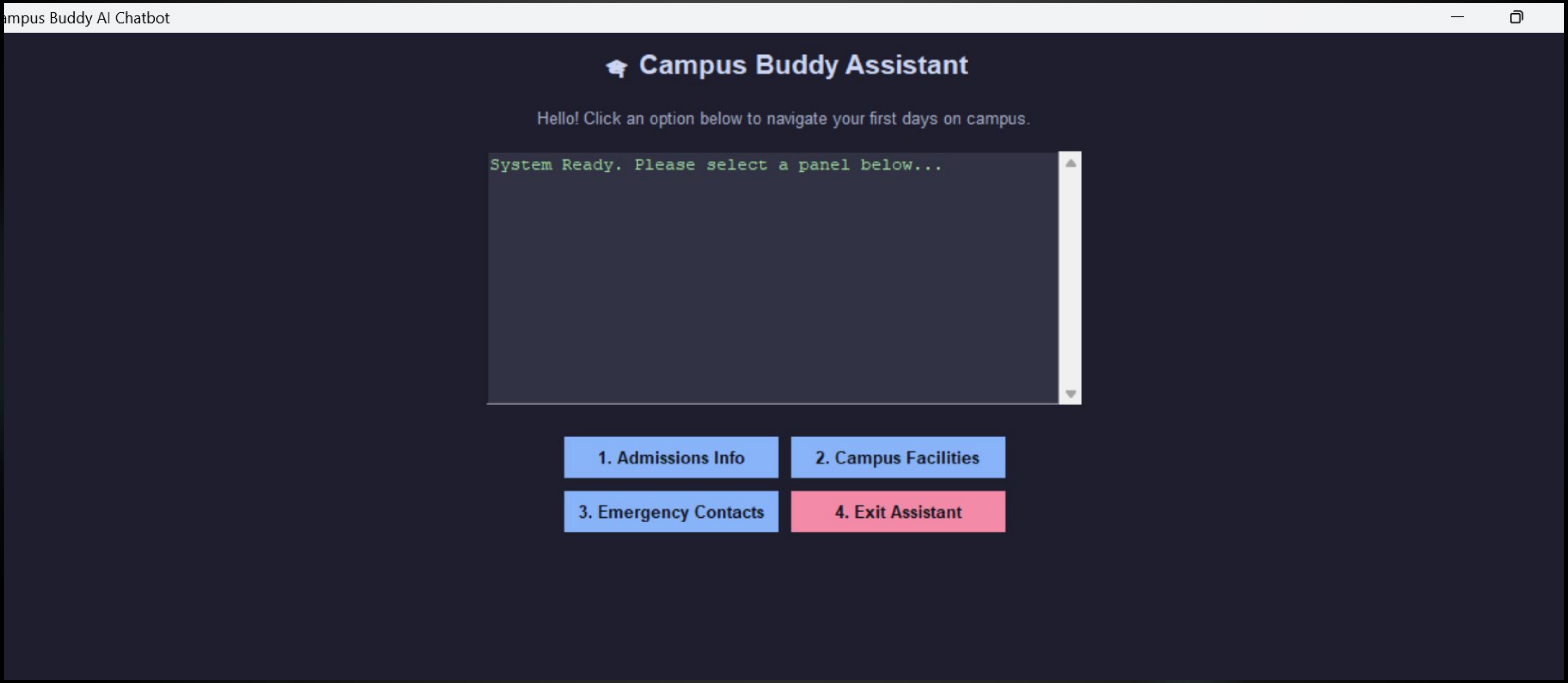

CodeImage

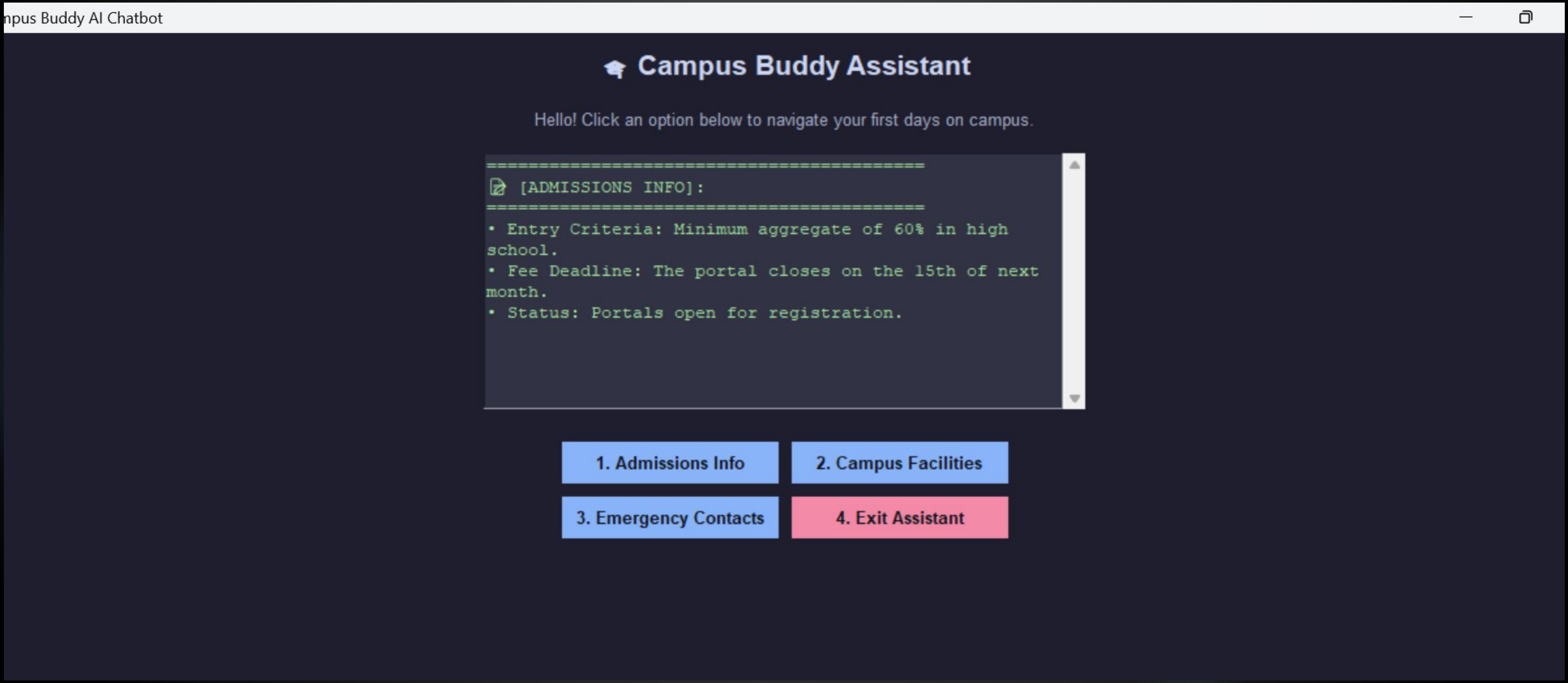
4. Screenshots with Explanation

- 2. Output 2.1



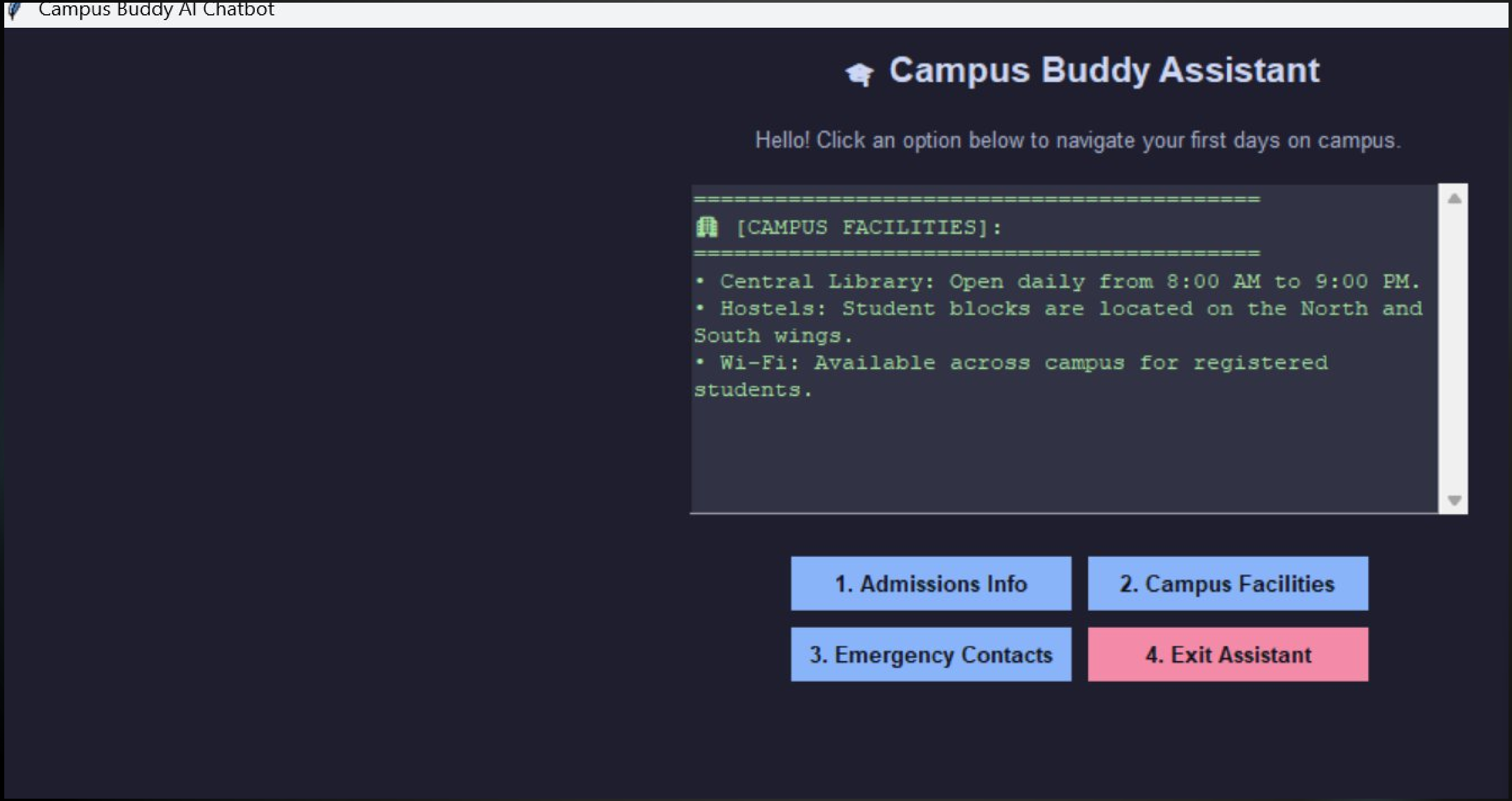
4. Screenshots with Explanation

- 2. Output 2.2



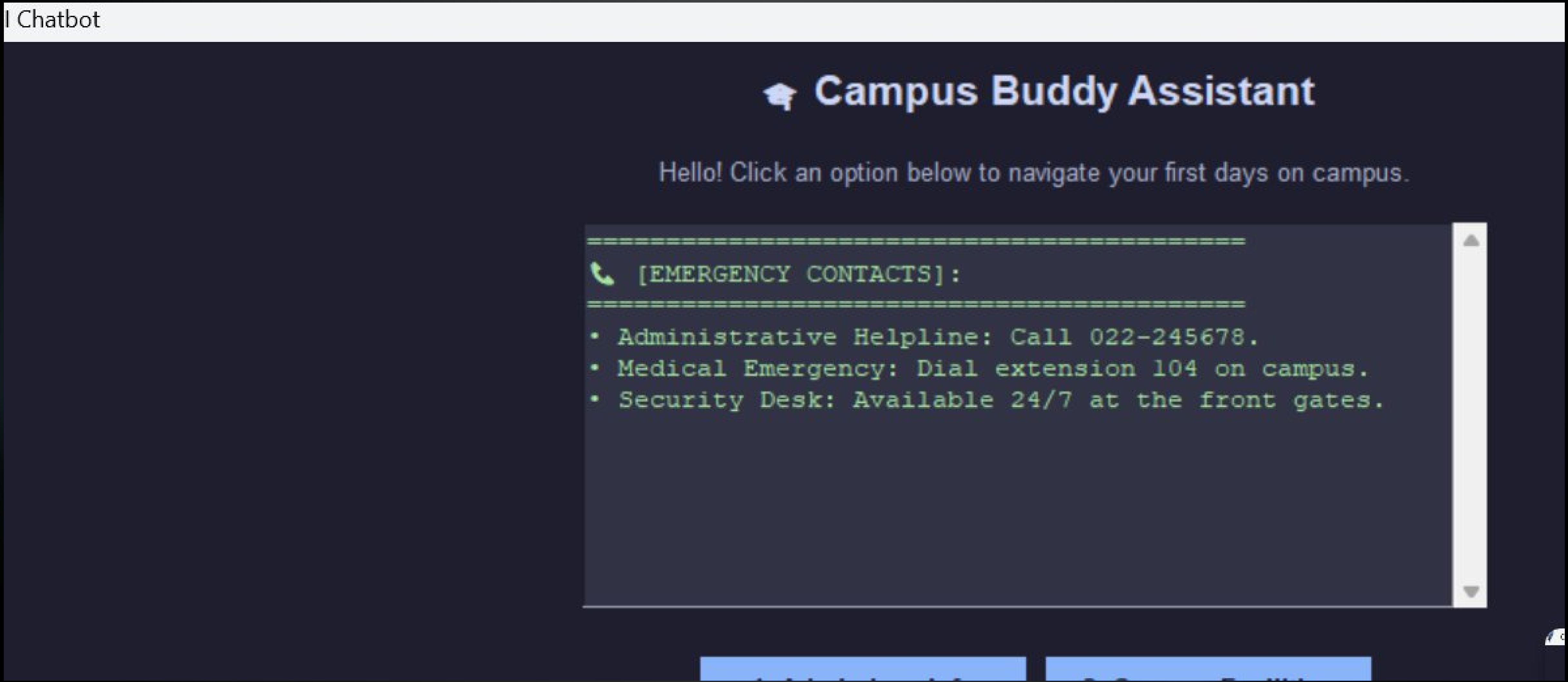
4. Screenshots with Explanation

- 2. Output 2.3



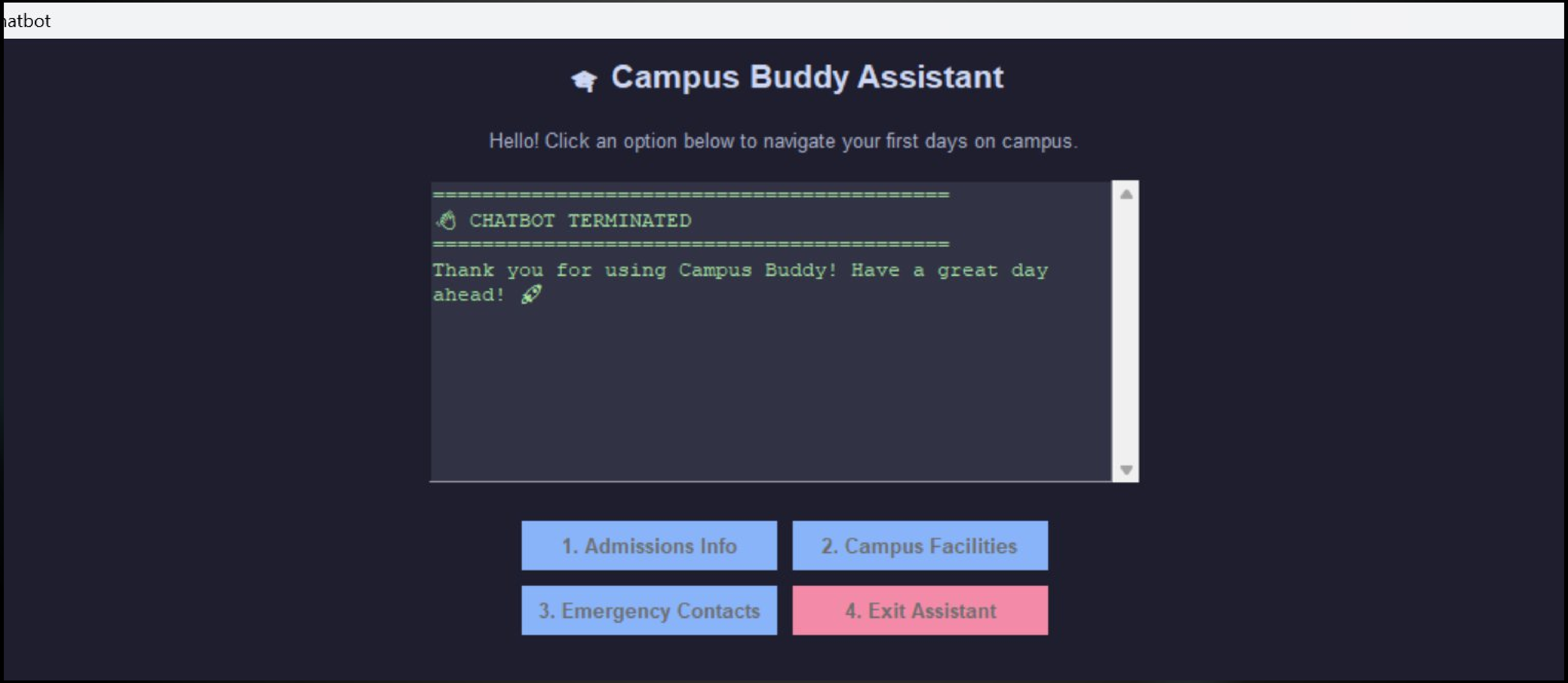
4. Screenshots with Explanation

- 2. Output 2.4



4. Screenshots with Explanation

- 2. Output 2.5



5. Project Links

- **Github Link:** https://github.com/ranejai954/lenovo-ai-internship-prerequisite-milestone-5-assignment-chatbot
- **Development Platform:** Visual Studio Code IDE Local Environment.

6. Conclusion & Summary

- **Project Outcomes:** The project successfully demonstrates the creation of a working desktop graphical user interface (GUI) application that tracks user clicks and returns appropriate contextual statements. The chatbot handles layout allocation frames cleanly and avoids loop crashes by handling user exits gracefully.

- Key Learnings:

1. **GUI Frameworks:** Learned how to create visual elements, window geometry sizes, and layout frames using Python's native Tkinter tools.
2. **State Control Logic:** Mastered event-driven programming, learning how to change on-screen components immediately when a user triggers buttons.
3. **Interface Accessibility:** Discovered that a button-driven GUI provides a much cleaner, error-free user experience compared to an open-ended terminal command box.